



# Routing Problems with Electric and Autonomous Vehicles: Review and Potential for Future Research

Themistoklis Stamadianos<sup>1</sup> · Nikolaos A. Kyriakakis<sup>1</sup> · Magdalene Marinaki<sup>1</sup> · Yannis Marinakis<sup>1</sup>

Received: 19 May 2022 / Accepted: 15 May 2023 / Published online: 31 May 2023  
© The Author(s) 2023

## Abstract

The transportation sector has undergone a major transformation in the past few years with the shift to electric mobility and the introduction of new, promising types of vehicles. Sustainability is the driving force of this revolution, but, these changes are expected to greatly impact the space of logistics operations. Electric vans have been in the market for a few years already, and they are comparable to gas-powered vehicles in certain applications; however, they are not the only ones with great potential. Drones and ground robots are two new types of vehicles, the characteristics of which offer remarkable opportunities in supply chains. Nonetheless, theoretical research on logistics operations with the abovementioned vehicles has been distant from reality. This research aims to help researchers explore the untapped potential of electric vehicles. To achieve this, a thorough look into their technical aspects is provided, to determine the key elements that distinguish them, make a comparison to the existing literature, and identify the research gap. Due to the increased complexity and the sensitivity of these vehicles to externalities and uncertainties in general, research should address and explore four major elements of these novel supply chains, *energy consumption, new vehicle types, dynamic environment, and communication* between vehicles.

**Keywords** Electric vehicles · Drones · Vehicle routing problem · Robots

---

✉ Yannis Marinakis  
marinakis@ergasya.tuc.gr

Themistoklis Stamadianos  
tstamadianos@tuc.gr

Nikolaos A. Kyriakakis  
nkyriakakis@tuc.gr

Magdalene Marinaki  
magda@dssl.tuc.gr

<sup>1</sup> School of Production Engineering and Management, Technical University of Crete, University Campus, Akrotiri 73100, Greece

## 1 Introduction

Modern means of transportation laid the foundation of today's society. Transportation vastly aided the development of commercial activities and the development of the economy and raised the standard of living of many people around the world. What makes transportation effective and efficient is proper planning, which can be split to two parts, transportation infrastructure and how it is utilized. Infrastructure consists of the networks that connect places, i.e., roads, and supporting utilities such as gas stations. How the above are utilized is what determines the efficiency of transportation.

The research of Dantzig and Ramser [1] was the first on this issue, presenting the truck dispatching problem and aiming to find the best possible way to deliver fuel to gas stations. Today, we refer to such problems as vehicle routing problems (VRPs). Since its inception, countless variants of VRP, and even more solution methods, have been presented [2]. By the end of the 1990s, most of the algorithms that are still used today for VRP were already introduced. Despite that, interest in VRP has yet to seize, since the means of transportation and the needs of people keep evolving over time. The latest addition to VRP, which initially emerged out of necessity but has proven to have additional benefits, is the employment of alternative fuel vehicles, namely electric vehicles (EVs) [3].

EVs have become a prevalent topic of research, both from an operational and a managerial viewpoint. The electric vehicle routing problem (EVRP) has gained a lot of popularity since the presentation of its first variant in [4]. Although some external parameters that affect the driving range of EVs have been addressed [5], they often use arbitrary values that do not reflect the conditions encountered in practice by EV drivers and do not represent reality. The objective of this paper is to address this gap by providing researchers with valuable information and future research directions that bridge the gap between theoretical research and the actual characteristics and abilities of all EVs. This will enable more accurate and effective solutions to be developed for the EVRP, which is crucial for promoting the wider adoption of EVs, reducing their environmental impact, and getting the most value out of them.

### 1.1 Sustainability and Electric Vehicles

Sustainable transportation is a global goal in the field of logistics and the use of EVs is one of the methods that will assist in achieving this goal. By prioritizing sustainability and making it a core part of the business strategy, logistics companies can help to mitigate the negative impacts of transportation on the environment and promote long-term economic and social sustainability [6]. By adopting sustainable practices and behaviors, logistics companies can help to create a more sustainable and efficient supply chain that benefits all stakeholders, including the environment, consumers, and the economy [7].

The EV space contains a plethora of vehicles, of different kinds and with different technologies. The types of vehicles solely equipped with electric powertrains

can range from two-wheeled electric-assisted pedal bikes to full-fledged electric trucks for logistics operations [8]. An area of high interest has been the introduction of autonomous vehicles in supply chains, such as drones [9–11] and ground robots. These vehicles are Internet-of-Things (IoT) devices, which may effortlessly communicate and coordinate with each other, revolutionizing logistics operations as we know them [12–14].

Each type of vehicle has its own unique advantages and limitations, and thus, should be utilized accordingly within supply chains. For example, flying drones are not bounded by traffic congestion, but have a low payload capacity. In contrast, ground robots have the ability to carry heavier items, but are bound to the road network. In combination, the strengths of each vehicle type can be leveraged as required. Overall, by correctly integrating these novel vehicles, supply chains become more effective and efficient, both financially and environmentally. Their autonomous nature and their ability to cooperate as a swarm has the potential to make operations more agile, and adaptive, with significant improvements in the quality of service.

## 1.2 Related Research

Given the popularity on the subject of EVs, there have been recent publications reviewing and discussing their application in VRP [5, 15]. Other publications on EV adoption [16–18] discuss the topic from a managerial and strategic standpoint while this research provides an in-depth analysis of the technical characteristics that affect the integration of electric-powered vehicles in the supply chain.

Drones and robots on the other hand belong to the class of autonomous vehicles. While they can be manually operated, when used in logistics operations, they are expected to be fully autonomous. Since fully autonomous drones are already in use for commercial deliveries; it is not a far-fetched assumption. In [19], the authors explore different classification criteria for drone routing problems and path planning problems and provide an overview of existing literature on the topic. A broader look into the potential use cases for drones was presented in [20], with both social and technical impacts being discussed. In [21], the need for advanced data management and policy frameworks during the transition phase is discussed, while in [22] the focus of the study is on fully autonomous vehicles and their impact on private and public transportation. In [23], the use of autonomous mobile robots for intra-logistics applications is discussed. Another relevant review is the one presented in [24], which discusses both drones and robots in two-echelon applications, however, once again from the traveling salesman problem (TSP) and VRP perspective.

In contrast to these publications, this research is not focused on reviewing the existing literature on the EVRP. Instead, the purpose of this study is to provide an overview of the current approaches in the literature on the subject of routing battery EVs, along with literature presenting EV experiments in the real world. This is motivated by the gap between the practical experiments and the computational studies. The authors aim to provide both a good understanding of the unique characteristics and the technical aspects of the new types of vehicles, to present existing approaches

in the literature, to identify important elements to be explored and addressed in future research, and to provoke new ideas and concepts that help with their adoption.

### 1.3 Research Methodology

The survey aims to highlight the discrepancy between the hard technical characteristics of EVs, including autonomous robotic EVs, and the approaches in the literature utilizing them and suggest future research directions. Therefore, the publications included are not limited to specific types of VRPs, but represent a wide spectrum of their applications and perspectives. The research is structured in the following way; first, the technical characteristics of EVs, and their published applications, are analyzed and discussed. Secondly, the research is expanded to include publications that utilize autonomous robotic vehicles, in tandem with EVs, extending the capabilities of both vehicle types.

The term EV can be used to describe a wide range of vehicles that use electric motors for propulsion, i.e., battery-powered EVs, hydrogen fuel-cell EVs, plug-in hybrids [25], and EVs with range-extending ICEs [26]. Each EV type has unique features that offer different advantages and disadvantages. This research is concerned with the most heavily constrained type, the battery-powered EVs.

To define the level of realism in EVRP research, a systematic review was conducted. Science Direct and Google Scholar were the two databases used to find relevant literature. The keywords used to find relevant research and review articles were “Electric Vehicle Routing Problem,” “EVRP,” “Vehicle Routing Problem with Drones,” and “Vehicle Routing Problem with Robots.”

The recently published review articles on EVRP [5, 15] were thoroughly studied to determine the current trends and the missing link between theoretical studies and real-world applications. The research papers included in EVRP reviews are mostly variants of VRP updated to employ EVs instead of traditional vans and trucks. It was found that although all review papers discuss the need for a more realistic representation of the real world, they do not mention any of the existing work regarding EVs outside of the theoretical research on EVRP.

Retrieving and reviewing all available research articles would be an impossible task; therefore, the research papers on EVRP that focus on parameters of interest were selected to be included in this paper, as parameters of interest are considered those that can be measured or predicted within a certain degree of accuracy, and used to calculate the energy consumption. Besides EVRP, general research regarding EVs was also considered. The research papers regarding EVs that were of interest are those that carried out tests, either in a lab environment or in real-world conditions, as well as papers discussing the viability of transitioning to purely electric vehicles or electric vans and trucks in case of commercial and industrial use cases. These papers were acquired by targeted research on the same platforms as before.

Charging infrastructure and its effect on EVs is also discussed; however, infrastructure development is not discussed as it is a separate issue from a technical standpoint. Nonetheless, charging technology and the different charging behavior of different EVs is showcased with the intent of highlighting the importance of

case-specific problem-solving for EVRP. Indicative EVRP research papers that have considered different charging technologies have been included in the review.

The contribution of this research can be summarized to the three following points:

- To present the range of VRP variants found in the literature utilizing these novel means of transport, describe the technical characteristics of those and their individual strengths and weaknesses, and explore the different ways they have been combined so far.
- To highlight the limitations of the recent VRP approaches found in the literature, the oversimplification over real-life applications, their static nature, and lack of uncertainty which would make them inapplicable in practice.
- To suggest a direction towards which the VRP research should focus, based on the current state-of-the-art means and their potential as IoT devices within a network of interconnected cooperative vehicles.

The structure of this research paper is the following: Sect. 2 provides the state of the art of electric vehicles and their technology, an assessment of the level of realism in the EVRPs, and raise concern on the related issues. In Section 3, technical information, strengths, weaknesses, opportunities, and threats, along with key literature references are provided for drones and robots. Section 4 raises questions based on the findings of the study and proposes future research directions. Finally, Sect. 5 presents the conclusions of the research.

## 2 Electric Vehicles and Logistics

Initially, this section provides a very short reference to the history of EVs, along with a comprehensive list of their most important strengths and weaknesses. An assessment of the realism level of the EVRP literature follows, and information regarding the energy consumption and charging, and parameters that affect them are discussed. A short technical overview of the currently available electric vans is presented as well.

### 2.1 Short History of Electric Vehicles

In the early 1900s, out of the 4200 vehicles registered in the USA, about 40% of which were steam-powered, another 40% was electric-powered, and the rest were gasoline-powered [27]. EVs were easier to operate compared to the gasoline-powered vehicles of the time. The Electric Vehicle Company was the biggest automotive manufacturer of the time, in the USA. Their business model was based on renting the vehicles for the day and taking them back for overnight charging and any necessary maintenance, a business model that has started once again to gain popularity. The company fell into some legal trouble and eventually went out of business. At the same time, the Ford Motor Company started the production of the highly successful Model T, which led

to the demise of the EV once and for all [28]. In the 1990s, General Motors allegedly self-sabotaged their own EV attempt, in favor of keeping their business model intact. The first commercially successful EV arrived in 2012 by Tesla. It was the first EV that could directly compete with the internal combustion engine (ICE) vehicles in terms of driving range and was proof that EVs can be an option [29].

## 2.2 Strengths and Weaknesses of Electric Vehicles

A description of the main strengths and weaknesses of EVs is given in the following list. They are not presented in any particular order.

- Strengths:
  - Running costs: The cost per unit of distance is usually lower for EVs compared to ICE vehicles, since electricity will in general be cheaper than gasoline or diesel [30].
  - Maintenance costs: EVs are mechanically simple compared to ICE vehicles. The average electric powertrain contains just a fraction of parts compared to modern ICEs [31].
  - Preferable for urban logistics: Given their zero tailpipe emissions and the lack of noise, EVs are perfect for urban applications [32].
  - Incentives: There are currently incentives in many countries across the world for potential EV buyers, such as tax reliefs, subsidization, and other [33].
- Weaknesses:
  - Range: It is one of the first points of worry for anyone looking to buy one. In the case of electric vans, Mercedes-Benz has published that after analyzing 1.6 million EV trips, almost all no longer than 100km [34], which is within the capabilities of the currently available vans.
  - Payload: The payload of EVs is a parameter that has a direct effect on range. Any vehicle with a payload will require extra energy to move compared to it being empty, given the extra inertia. It should be noted that EVs usually have a lower maximum payload rating compared to ICE vans [35].
  - Charging: The speed of charging and the available infrastructure is another point of worry. Charging stations and especially fast-chargers are not so common, yet [36].
  - Cost: While some costs associated with EVs are lower compared to ICE vehicles, the initial cost of purchase is more often noticeably higher than that of conventional ICE vehicles [37].

## 2.3 Assessing the Level of Realism in EVRPs

This section aims to compare the real world to the existing literature on EVRP. Charging and discharging are the main issues to be discussed, as they are very important for a realistic VRP. There are a few critical technical aspects of EVs that are usually not considered in EVRP studies and affect both charging and discharging.

In practice, many parameters must be concurrently addressed, many of which are interdependent. The most recognizable are vehicle payload and speed, temperature, use of auxiliary devices (i.e., air conditioning), driving characteristics, the current state of charge (SoC), and energy losses (drag, tarmac friction, etc.).

Some of these have been studied both in real life and in the EVRP literature, and are presented in the following paragraphs.

### 2.3.1 Temperature

Temperature is a very influential factor of EV general operation. For example, using an EV in colder climates has an effect on the efficiency of the battery, while it necessitates the use of heating elements for the occupants of the cabin, leading to further energy depletion. The optimal battery operating temperature typically ranges from 22 to 25 °C, depending on the vehicle [38]. During the summer, charging can become slower as a result if the vehicle cannot maintain a low enough temperature during charging. Temperature does not affect battery performance only during charging, it is highly important during discharging too, especially when vehicles are parked outside, and exposed to the elements. Cold operating conditions can also have an effect on battery aging [39]. Normal driving does not really heat up the battery of the EV; therefore, extra measures do not need to be taken to bring the batteries to operating temperature. In [40], a graph of energy consumption throughout the year, as temperatures change, shows the effect temperature had on the range of a Nissan Leaf EV. Furthermore, using the EV within city limits at temperatures lower than 15 °C resulted in a poor range. In [41], the authors focused on battery degradation for lithium iron phosphate batteries. One of the tests they performed showcased the battery capacity loss against the charging cycles for three different temperatures, for three different depths of discharge. While the tests were performed in lab conditions and not used in actual vehicles, the results might not exactly coincide with real-world tests, but, they prove that both the depth of discharge and the operating temperature may affect the longevity of batteries. In reality, the depth of discharge may not be an overall issue, given the battery controller spreads the demand across all battery cells.

### 2.3.2 State of Charge

The SoC of the vehicle is a very critical parameter for charging as well. The demand profile of an EV battery was studied in [42]. The results of their tests suggest that the SoC should be in the range of 20% to at most 90%. Their validation experiment showed a linearity in battery voltage between 20 and 80% for a constant current value, both during charging and discharging. In [43], the researchers focused on finding the best charging scenarios for EVs. They also presented a lot of relevant literature on the subject of EV charging. They suggest a similar SoC window of operation for EVs, between 20 and 80%.

### 2.3.3 Consideration of Realistic Parameters in EVRPs

Most of the EVRP literature, especially earlier studies, used very simplified, linear energy consumption models. Later studies started to introduce elements of realism in their models, i.e., weather conditions or vehicle speed; however, there still is a lot of room for improvement.

The energy consumption functions used in the literature have not been related to any existing vehicles, with the exception of those that use some characteristics of passenger EVs. The problem with that tactic is that the most important factor, weight, is not accounted for. Some noteworthy studies that tend towards a realistic energy consumption estimation are the following. An EVRP considering the effect that vehicle load would have on the battery was carried on real-life data from Austin, TX [44]. In [45], a new formula was developed for energy consumption, taking into account speed and weight. They tested the performance of their hybrid genetic algorithm on a Beijing road network. An EVRP with time windows (EVRPTW) was solved in [46], considering the energy consumption rate as a function of speed and weight. In [47], the EVRP was solved using an ant colony optimization algorithm and an adaptive large neighborhood search (ALNS) algorithm, aiming to minimize energy consumption instead of distance traveled, to prove it is a superior tactic. To calculate the energy, the weight, speed, and road friction were considered. In [48], an energy consumption model that considers the topography and the speed profiles was developed. First, the road network is evaluated and, then, the two-stage approach seeks the solution. The experiments were carried out on a Swedish road network. The effect that ambient temperature could have on vehicle routing was considered in [49], given the need for a tolerable cabin temperature and the fact that EVs are less efficient in cold environments. Their objectives were the minimization of the fleet size and the minimization of total energy consumption. They used passenger EVs and, therefore, there may be inaccuracies when compared to electric vans. In [5], a very detailed literature review was presented along with a new model for EVRPTW. The energy consumption rate of the new formulation is the richest to date, accounting for aerodynamic, tire, drivetrain, ancillary, and other energy losses. Another recent addition is the work presented in [50]. The authors presented a problem in which the distance the vehicles have to travel is not too long; subsequently, charging is not an issue. An extensive analysis was carried out on case studies of four cities to demonstrate the sensitivity of the parameters of the problem. These parameters were the capacity, the max travel time, the service time, and the range of the EVs.

### 2.3.4 Charging Infrastructure

The terminology for charging stations is electric vehicle supply equipment; however, in this research, they are referred to as chargers or charging stations, since they are commonly referred to as such.

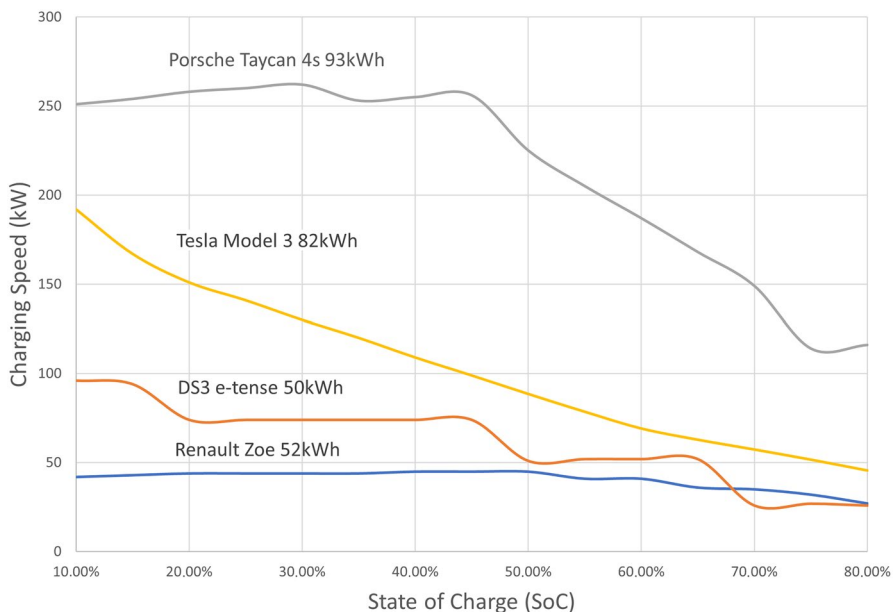
Currently, no universal standards exist on charging speeds or charging ports. Charging can be categorized into two types, alternating current (AC) charging and directional current (DC) charging. AC is typically used for most traditional slow



chargers, while fast chargers are more often than not of the DC type. DC charging bypasses the AC-to-DC converter of the vehicle and directly charges the battery, making charging even more efficient. Each current type may be divided into two levels, depending on the region. North America, Europe, China, and Japan have adopted four different charging port standards. Further insights into the charging speeds and technology standards are provided in [51] and in [52].

There are two different aspects of charging to consider. The first one is whether charging is full or partial. As stated previously, the SoC has to be maintained below a certain percentage to extend the life of the battery. Partial recharging (PR) has already been implemented and suggested by many researchers in the field of EVRP. In [53], four different cases of EVRPTW were solved, each with a different charging limitation, and concluded that allowing PR, as many times as necessary, is the best approach. [54] solved the EVRP with PR while considering charging parameters such as time-dependent energy pricing and the efficiency of the EV's energy converter. In [55], a three-phase mat-heuristic was introduced for the time-effective EVRP with PR, aiming to minimize the number of EVs used and the total time. In [56], the authors considered the limited charging capacity of charging stations and explored PR strategies to improve charging times. In [57], an EVRPTW was solved, showing the advantages of quick charging in route planning compared to a single charging policy that leads to a reduction in cost and fleet size.

The other aspect is the charging profile. Each EV has different components, which means that the charging characteristics are not universally the same. In Fig. 1, the charging profiles of four EVs are presented to showcase the diversity of charging profiles among EVs. The Tesla Model 3 and the Renault Zoe are the only two



**Fig. 1** Examples of different charging curves

vehicles with a charging curve that has an almost linear behavior. On the other hand, the DS3 e-tense charging curve resembles a stepped line graph and the Porsche Taycan 4 s charged at about 250kW up to 45% and then the charging speed linearly drops until about 75%. Charging curves change significantly outside of the bounds of the graphs, meaning before 10% and after 80% charging takes place slower to avoid damaging the components of the EV. In literature, non-linear charging profiles have already been proposed. An EVRPTW was solved in [58], utilizing a different model and a concave non-linear charging function with the objective of minimizing the total operational costs. In [59], a nonlinear charging function was considered along with limited capacity charging stations when solving EVRP. A two-layer genetic algorithm was presented in [60] for solving EVRPTW with multiple depots and partial, non-linear recharging. In [61], a lot of attention was paid to the battery and the effect that the depth of discharge would have on the life of the battery. They suggest maintaining a high SoC, as expected. The issue with the non-linear functions currently found in literature is that they do not represent any existing vehicles.

### 2.3.5 Fast Charging

Being able to fast-charge a vehicle is critical in logistics operations [62]. Refueling for ICE vehicles is never thought of as an issue, as there are plenty of gas stations that can quickly refill the tank that will suffice for all-day use, on most occasions. While in [63] it is suggested that fast charging will lead to battery degradation, fast charging is in many cases inevitable. Fast charging makes the battery heat up due to the quick surge of energy. Manufacturers are aware of the related issues and in most cases, vehicles equipped with fast-charging capabilities are, also, equipped with battery liquid cooling or other technologies that help minimize the long-term side effects of extreme temperatures.

Fast charging has not been researched extensively in EVRP, yet [57, 64, 65]. Nonetheless, all research indicates the benefits of fast charging on logistics operations.

### 2.3.6 Battery Swapping

Battery swaps have also been researched, but they are not generally applicable to electric vans, as their large batteries cannot be swapped easily. In [66], the EVRPTW with synchronized mobile battery swapping (SMBS) was introduced. An EV can request a battery swap on the fly. A battery swap vehicle drives to the meeting point the driver of the EV requested, to make the battery swap. This method of replenishing energy is proposed to alleviate range anxiety. They tested SMBS against traditional charging stations, which they estimate to cost about 3.9 times more. A two-echelon EVRP but with battery swapping stations instead of traditional charging stations was solved in [67]. In [68], a variant of EVRP was presented, combining recharging and battery swapping. More specifically, EVs can recharge during their trips or swap their batteries in specific locations, with the help of battery-swapping vans, meaning two VRPs have to be solved at the same time.

## 2.4 Technical Overview of Existing Electric Vans

This section aims to provide researchers with a basic overview of electric vans and purpose-built electric vehicles for urban logistics applications. Most of the instances used in EVRP are several decades old. The purpose they serve is algorithm comparisons. There is a need to create instances of realistic characteristics that can provide insights for real applications. To help researchers achieve that, basic information, such as the vehicle range and how it is measured, payload capacity, and charging speeds for existing vehicles used currently for urban deliveries, are presented.

Battery technology is the most important factor impacting the adoption of EVs. Vast amounts of resources are devoted to research and development. Modern EV batteries are of the lithium-ion type [69]. A lot of small-size batteries are glued together in series or in parallel and together they form the battery of the vehicle. In the case of a modern Tesla, more than 7000 small batteries are used. Their energy capacity is measured in kilowatt-hours (kWh).

EV efficiency from tank to wheel, meaning from the battery, through the electric motor, to the wheels is very high. ICE vehicles have a value ranging from 0% when idle to a maximum of 20% at the optimal engine rotation speed (rounds per minute or rpm) and operating conditions. EVs can achieve a tank-to-wheel efficiency of more than 90% regardless of the speed of the vehicle [70]. This is partly owed to the lack of the ICE and partly due to the simple, single-gear, transmission mechanics.

On the other hand, contemporary EV batteries have not yet achieved the same energy density as petroleum products. More specifically, diesel fuel has a specific energy value of 45MJ/kg, and one of the most popular EVs, the Tesla Model 3, has less than 0.5MJ/kg. Therefore, to achieve a usable driving range, battery packs tend to be heavy, accounting for as much as a quarter of the total vehicle weight. For freight vehicles, this is especially bad, since a heavier battery pack lowers the maximum payload capacity of the vehicle.

Electric van battery capacities may range from 37.3kWh for a VW ABT e-Transporter, able to provide 132km of driving range according to the Worldwide Harmonized Light Vehicle Test Procedure (WLTP). On the other end of the spectrum, the Toyota Proace Electric may be equipped with a 75-kWh battery providing an extra 200km of range compared to the Volkswagen, at 330km, according to WLTP. In the WLTP, the vehicles are tested with 15% of their rated payload. Still, according to manufacturers and researchers, in most cases, vans do not cover distances greater than the WLTP range of the electric vans found today on the market [34, 71].

Table 1 presents a few electric vans appropriate for different kinds of logistics operations. They range from commercial hatchbacks, like the Renault Zoe Commercial, to the electric variant of the Sprinter from Mercedes-Benz. The table contains the model names, the battery capacity in kilowatt-hours, the range in kilometers, the maximum payload in kilograms, and cargo volume in cubic meters, along with the estimates of the manufacturers for the charging time given a charging station able to provide the maximum single-phase AC charging by European standards. In some cases, more than one battery option was available. An odd feature is that the larger Fiat E-Ducato and Mercedes eSprinter when equipped with the large battery pack option have a lower maximum payload. This is necessary in order to keep the gross

**Table 1** Technical specifications of modern electric vans

| Model                        | Battery<br>(kWh) | Range<br>(km) | Payload<br>(kg) | Cargo<br>(m <sup>3</sup> ) | Charging<br>(kW)/(hours) |
|------------------------------|------------------|---------------|-----------------|----------------------------|--------------------------|
| Renault Zoe Commercial       | 52               | 394           | 475             | 1                          | 7/9–10                   |
| Renault Kangoo ZE 33         | 33               | 240           | 650             | 3.5                        | 7/6                      |
| Toyota Proace City (SWB)     | 50               | 273           | 803             | 3.8                        | 7/7.1                    |
| Nissan e-NV200               | 40               | 200           | 705             | 4.2                        | 6.6/7.5                  |
| Maxus e-Deliver 3 (SWB)      | (35/52.5)        | (159/243)     | (865/945)       | 4.8                        | 7.2/(6/8)                |
| Toyota Proace Electric       | (50/75)          | (229/330)     | 1226            | 5.8                        | 7.4/(7.5/11)             |
| Mercedes eVito (L2)          | 41               | 148           | 905             | 6                          | 7.2/6                    |
| Volkswagen Abt e-Transporter | 37.3             | 132           | 996             | 6.7                        | 7.2/5.5                  |
| Fiat E-Ducato                | (47/79)          | (174/283)     | 1160/845        | 10                         | 7/(8/-)                  |
| Mercedes eSprinter           | (41/55)          | (120/168)     | 1045/891        | 11                         | 7.4/(6/-)                |

weight below 3.5 metric tons of weight, given that larger batteries weigh more. The reported range values are the WLTP combined use (55% urban and 45% extra-urban) test results.

The WLTP test is carried out under controlled laboratory conditions and consists of four different phases, including low- and high-speed driving, and a combined cycle test that simulates urban and highway driving conditions. The test vehicle is equipped with a standard set of sensors and measuring devices. The test is performed according to a standardized driving profile, which takes into account various factors such as vehicle weight, engine power, and aerodynamic resistance. According to the WLTP technical regulations, the temperature is set at 23 °C and 14 °C. The type, pressure, size, and condition of the tires used in the test are also monitored since tires are a critical factor in energy consumption. One of the missing elements is wind speed, which could affect the range of an EV when traveling at highway speeds. The exact details on the procedure followed can be found in [72].

It is evident that the best-valued vehicles are small- and medium-sized vehicles (based on the cargo volume), depending on the use case. However, it could be argued that unless the weather conditions are harsh, vehicles such as the Renault Zoe Commercial could be replaced by a trike or other, more energy-efficient vehicles compared to the Zoe. Furthermore, the two largest vehicles would be suitable for applications where light items of big dimensions have to be transported.

In contrast to the vehicles of Table 1, new companies and start-ups that have focused on EVs strictly developed for urban logistics are even more interesting. While bigger companies have already invested millions in research, development, and tooling for ICE vehicles and try to adapt them to electric power plants to maintain their economies of scale, small companies produce purpose-built vehicles from scratch that do not have to fit into any existing standards. Offerings from Melex (Poland), ALKE (Italy), Paxster (Norway), and others have presented many vehicles in this segment. Table 2 presents the technical specification for some of these vehicles. The model, battery capacity, range, max payload, cargo volume, and max speed are reported for each vehicle according to the manufacturers' websites. The Paxster

**Table 2** Technical specifications of small electric vans only for urban use

| Model            | Battery capacity<br>(kWh) | Range<br>(km) | Max payload<br>(kg) | Cargo volume<br>( $m^3$ ) | Max speed<br>(km/h) |
|------------------|---------------------------|---------------|---------------------|---------------------------|---------------------|
| Paxster L6e      | 9.2                       | 141           | 240                 | 0.6+0.8                   | 45                  |
| Melex 3 (391.1H) | 7.2                       | 59            | 620                 | 2.6                       | 43                  |
| ALKE ATX 320E    | 10                        | 72            | 610                 | 2.8                       | 44                  |
| ALKE ATX 340E    | 20                        | 200           | 1185                | 2.8                       | 44                  |

L6e has two cargo compartments, one in the front and one in the back. The Melex 3 and the ALKE ATX320E are quite similar, while the ALKE ATX340E offers a greater battery size and greater payload and range.

### 3 Drones and Robots in Logistics

The current limitations of EVs, such as their range restrictions and dependence on charging infrastructure, pose significant challenges to their widespread deployment. One of the best ways to enhance the reach of EVs without the need for larger batteries, frequent visits to charging stations, or the need for a breakthrough in battery technology, is to combine them with autonomous vehicles such as drones and robots, offering a promising solution to the current limitations of EVs.

Furthermore, this integration has the potential to unlock new and innovative use cases for EVs, providing additional value to stakeholders, such as fleet operators, consumers, and the environment. In this light, the combination of EVs with drones and robots is a promising area for future research and development, with significant potential for advancing the deployment and utilization of EVs in a variety of domains.

The following subsections provide a comprehensive view of the various research paths already discovered within the scope of VRP, along with some indicative examples from real drones and robots. Although many researchers have opted to use them independently, in the context of deliveries, the most realistic application is the use of EVs as mobile depots for these vehicles.

One of the most popular means of transporting parcels in the recent VRP literature is drones. Their popularity rose in the past 5 years, since well-known commerce and logistic companies, such as Amazon and DHL, openly started considering them for last-mile deliveries, in an autonomous way.

Drones inherit the environmental benefits of all-electric vehicles, such as the lack of local emissions, but also their drawbacks which is the limited range of operation without requiring recharging. Their maximum payload is another limitation, as the energy required to lift heavy items would make them practically unusable. Moreover, drone operations are very sensitive to weather conditions, as unpredictable factors like wind and rain greatly affect them. Nevertheless, unlike EVs, drones have the ability to overcome obstacles, such as traffic and buildings, making deliveries

faster in principle by traveling shorter distances to reach a destination. In [10], it is suggested that drone and ground vehicle integration in logistics can offer multiple benefits including monetary and environmental.

### 3.1 Technical Overview of Drones

This subsection discusses the technical characteristics of drones. Table 3 displays the specifications found in the corresponding official websites or estimations of third-party online sources in cases where official values are not publicly available for the most well-known service providers.

Real-life applications are more common than one might think and expand among many different areas. Amazon has been an early adopter of drone deliveries, developing their own drones, capable of carrying payloads of up to 2.25 kg, for trips of 30 min at maximum.

In humanitarian applications, Matternet uses drones to transport test samples and medical necessities, while Zipline created a modular fixed-wing drone to be used in blood bag transportation, in Rwanda, providing significant benefits and saving lives.

Information regarding drone specifications is limited and the available measurements are neither standardized nor confirmed by independent third parties, thus, are not comparable to each other. Subsequently, these characteristics should be seen as rough estimates and their true capabilities remain proprietary knowledge of the companies that developed them.

Most drones are of similar size and have similar capabilities, as they are restricted by the battery weight. Two main variants may be distinguished, fixed-wing, and rotary-wing drones, the latter of which is the most common. In most countries, the main barrier to their use in commercial logistics applications is the strict regulatory framework.

The research presented in [77] is the only published research based on real data from drone flights, analyzing their energy expenditure and providing insights for their use. The drone used is a commercially available drone suitable for transporting items. The energy consumption was determined to be on average 0.08MJ/km, carrying a payload of 0.5kg. Despite the not so wide scope and the limited data used, it is a move in the right direction. The drones were tested against other modes of transportation as well, such as cargo bikes which were also proven to be a very good option in terms of energy expenditure.

**Table 3** Technical specifications of drones

| Drone model               | Type        | Range (km) | Payload (kg) |
|---------------------------|-------------|------------|--------------|
| Amazon Prime Air [73]     | Quad-copter | 16         | 2.3          |
| DHL Packetkopter 3.0 [74] | Tilt-wing   | 8.3        | 2            |
| DHL Packetkopter 4.0 [74] | Titl-rotor  | 65         | 4            |
| UPS Matternet M2 [75]     | Quad-copter | 22.5       | 2            |
| Zipline [76]              | Fixed-wing  | 101        | 1.8          |

### 3.2 VRPs with Drones

Research on the subject of VRP with drones (VRPD) and its variants is a very recent addition to the literature. The first problem of its kind utilizing drones was introduced in [78], more specifically the traveling salesman problem with a flying sidekick (FSTSP). In this simple integration of the two vehicle types, the drone is an assistant to the road vehicle. The FSTSP formulation was extended to include multiple TSP with drones [79]. A recent addition to the TSP with drones was presented in [80], with the speed of the drone being determined by the weight of the packages it is carrying, similar to the work presented in [81]. The first research to address the VRPD is [82], presenting the first formulation of the problem, assuming identical travel speeds for both types of vehicles, which does not realistically reflect the drones' capabilities. A maximum coverage problem with drones was presented in [83]. In [84], a generalized approach was proposed considering VRP with transportable resources, including drones. In [85], the authors presented a set-covering problem for instant deliveries and look for the best drone take-off locations. They minimized both the number of vehicles and the makespan. In [35], the novel electric VRP with drones was introduced, minimizing the energy consumption of both types of vehicles, while considering the weight.

There has been great diversity in the concepts and the scenarios researchers have presented. Routing two different types of vehicles bears many difficulties in terms of synchronization. The authors in [86] addressed separately the truck and drone path planning and then proceed to jointly optimize them. In [87], a movement synchronization VRP was proposed, applicable to other types of vehicles too. In [88], a more realistic formulation was presented as drones are transported by trucks (serving as depots and battery swapping stations) and can make multiple deliveries, allowed to return to any station, making many trips if necessary. In [89], the authors set to lower the customer waiting time. They allocated multiple drones per truck and conduct a case study. In [90], only drones were allowed to make deliveries to customers while the trucks transport them to the designated launch/retrieval locations. Both in [79] and in [91], two-echelon approaches in drone integration were proposed, using the trucks as mobile depots, with the latter allowing direct drone deliveries from the depot too. In [92], drones and trucks were routed independently, with drones making only a single delivery each time.

Just like EVs, there are a number of factors that may have an implication on the logistics operations with drones. In [93], the height of the delivery and its effects on the routing operation were considered. In [94], a more realistic drone delivery system was introduced, as it considered no-fly zones and wind conditions. They assume that rooftops of city buildings may be charging points. In [95], the problem emphasized weather impact and included collision avoidance as well.

Some applications that may benefit from the use of drones have also been proposed. A dynamic VRP for food delivery application was solved in [96]. In [97], the intent is to select the optimal hub for delivering essentials in disaster relief scenarios to best serve those in need and present a theoretical example.

Energy consumption is a very important parameter for drones as well. In [98], non-linear and linear energy functions were compared while also considering the weight parameter, giving emphasis on the energy consumption.

There are several review papers on drone routing problems in the literature. In [99], the authors highlight that routing problems with drone integration are an emerging branch of research. In [100], the review focuses on routing problems with drones, reviewing the existing TSPs and VRPs. [101] includes in their review a comprehensive list of the recent literature concerning drone integration in multiple TSPs. In the review paper [102], a drone problem taxonomy is proposed along with a discussion on practical applications. In [24], the drone integration problem is reviewed from a two-echelon perspective. They provide insights on issues encountered and concentrate on modeling perspectives. Finally, in [103], the factors that impede real-world applications are discussed and the related research gaps are highlighted.

Although drones are robots and often in literature they are considered as such, it is useful to address them separately in logistics operations given their different properties. In this paper, robots are considered the ground vehicle equivalent of drones.

Similarly to drones, the main application for robots in logistics has, for the greater part, been urban deliveries. Drones and robots share both advantages and disadvantages. They both have zero local emissions, can operate with some degree of autonomy, and are quite versatile, but, they both suffer from a limited operating range. Nonetheless, robots have some unique traits that cannot be matched by drones, like carrying heavy and sizeable items and the ability to operate under harsher conditions, since they operate on the ground and are less affected by wind, and rain. In contrast to drones, robots cannot make deliveries at altitude (i.e., balconies and rooftops) and are bound to follow roads or sidewalks (depending on the size of the robot), traveling longer distances compared to drones. Last and not least, operating on the ground makes them more vulnerable to stealing or other sabotaging efforts.

The VRP variants with robots found in the literature have many assumptions in common, as listed below.

- Instantaneous deployment and retrieval times
- Fixed operational time independent of the payload or speed
- A static, perfect-conditioned environment of operation.

Overall, a combination of drones and robots might bear the most benefits, as their joined strengths could outweigh their individual drawbacks.

### 3.3 Technical Overview of Robots

It is useful to provide a comprehensive presentation of the capabilities of the robots currently available and discuss their technical characteristics. Table 4 displays the values found in the corresponding official websites or estimations of third-party online sources in cases where official values are not publicly available. There are more robot vehicle models being developed and tested but since no specifications of any kind are available, they have been omitted.



**Table 4** Technical specifications of robots

| Robot model        | Size<br>(Lmm × Wmm × Hmm) | Weight<br>(kg) | Speed<br>(kph) | Range<br>(km) | Payload<br>(kg) |
|--------------------|---------------------------|----------------|----------------|---------------|-----------------|
| Amazon Scout [104] | 762 × 610 × 737           | 45             | 24             | -             | 23              |
| Starship [105]     | 678 × 569 × 1248          | 23             | 6              | 6             | 10              |
| TeleRetail [106]   | 1473 × 711 × 864          | 27             | 56             | -             | 35              |
| DeliRo [107]       | 962 × 664 × 1089          | -              | 6              | -             | 50              |
| KiwiBot [108]      | 559 × 432 × 559           | -              | 2.4            | 20            | -               |
| Serve [109]        | 790 × 640 × 1050          | 73             | 10             | -             | -               |
| Fedex Roxo [110]   | 914 × 711 × 1473          | 91             | 16             | -             | 45              |
| Nuro R2 [111]      | 2740 × 1100 × 1860        | 1150           | 40             | -             | 190             |

Robots come in many sizes, ranging from smaller ones intended only for sidewalk use to larger ones that could potentially use the streets. The wide range of available robots makes them a versatile vehicle type, able to meet the requirements of practical applications. To make the use of robots in logistics operations more common, two separate issues have to be addressed. First, the technologies related to their use must be improved, mostly with respect to autonomous operations, and secondly, laws and regulations regarding their use have to be set in place. In Fig. 2, four of the robots from Table 4 are displayed.

**Fig. 2** KiwiBot (top-left), Amazon Scout (top-right), Starship (bottom-left), Nuro R2 (bottom-right)

### 3.4 VRPs with Robots

From an academic perspective, the proposed VRPs with robots found in the literature can be grouped based on three criteria that emphasize their major differences. These criteria were chosen as these choices alter, not only certain constraints or the objective but the entire underlying practical application. The criteria are the following:

- Customer delivery vehicle: This determines whether only the robots (R) or both (B) types of vehicles are allowed to deliver the packages to the customer locations.
- Robot deployment: Robots may be deployed from designated locations (DL), full-fledged robot depots (RD) with storage/recharging capabilities, or only at customer locations (CL).
- Robot transportation: This characteristic indicates whether the robots are transported by the truck(s) to/from the deployment locations (Yes) or not (No).

#### 3.4.1 Customer Delivery Vehicle

Models that allow only robots to make deliveries to customers represent the use of robots in the last-mile delivery context. This is aligned with the efforts to reduce emissions, as the bigger multi-ton trucks do not have to move from customer to customer themselves. These models work great when delivering packages of insignificant weight; however, it proves problematic when there are packages that have to be delivered by the truck itself. In practice, this model cannot work in solidarity. Using both trucks and robots for deliveries is a far better option, not only for the mentioned reasons but when controlled substances or age-restricted items have to be transported.

Moreover, when employing both types of vehicles from deliveries, more flexible models that require a lesser degree of synchronization can be explored, depending on the available resources. A scenario of employing trucks as mobile robot depots which deploy robots at customer locations was the first to be proposed. Another scenario is the completely separate routing of trucks and robots, by having dedicated robot depots, a concept that does not require the same level of synchronization.

#### 3.4.2 Robot Deployment

The choice of robot deployment approach significantly affects both the complexity of the problem and its effectiveness. Using designated locations for robot deployment is a realistic and low-cost approach. Despite the lack of storage or charging options, robots may be deployed from such places while trucks continue on their route, and robots may return there at the end of their route and await retrieval. By developing designated robot depots, robots become less dependent on other vehicles, both in terms of synchronization and in terms of charging, since they would be able to recharge on their own. In case deliveries are performed only by robots (that are not moved by trucks), then the problem becomes two-echelon, meaning, there are two separate levels of routing. When customer locations are the only place

of possible deployment and retrieval, it is described as a mixed customer delivery approach. This application is arguably the easiest to implement since the robots are used as sidekicks to the trucks and there is no need for any kind of infrastructure; however, it would only be effective if there are multiple customers in close proximity.

### 3.4.3 Robot Transportation

Most approaches found in the literature use trucks to transfer both the parcels and the robots. The main advantage of this concept is that it does not require significant infrastructure. Thus, it would make financial sense to adopt this approach when operating in a suburban neighborhood, where robot depots capable of having their own robot fleet might not be cost-effective to maintain.

In contrast, when operating in a densely populated urban environment, having robot depots with permanent robot fleets will free important space in the truck which would otherwise be occupied by the robots. If a certain robot depot requires more robots, those could be transported from one to another, given that the robots used in that application can be transported by a van.

Table 5 provides a comprehensive summary of the various VRPs that incorporate robots, as reported in the literature. The first column lists the relevant publications, while the second column specifies the types of vehicles that make deliveries. The third column outlines whether or not the robots are transported by the EVs. The fourth column presents the delivery scheme of the robots' operations. The subsequent two columns indicate the objective of the studies and the availability of a mathematical model, if any. The final two columns present the solution approach adopted and the number of trucks utilized, respectively.

## 4 Future Research Directions

In recent years, the EVRP has gained significant attention due to the increasing interest in sustainable transportation. While several studies have addressed the EVRP, there are several potential extensions that could improve the problem's practical relevance. EVs are often marketed and sold as high-tech products, with a focus on their technological advancements, performance capabilities, and connectivity features. The growing trend of smart mobility and the Internet of Things (IoT) have further enhanced the tech components of EVs. By integrating EVs with connected services such as navigation, charging stations, and energy management systems, the vehicles become part of a broader ecosystem of connected devices and services. Furthermore, they are equipped with a variety of sensors and electronic components that generate a wealth of data, which can be analyzed to gain insights into vehicle performance, energy consumption, and user behavior. The availability of this data has the potential to revolutionize the way we understand and optimize the use of EVs.

A series of questions can be raised, which have yet to be answered by the ongoing research in the EVRP field:

**Table 5** Characteristics of published VRPs utilizing robots

| Reference | Cust. Del. Veh     | Robot Trans | Robot Depl | Objective       | Formulation | Approach                    | Trucks |
|-----------|--------------------|-------------|------------|-----------------|-------------|-----------------------------|--------|
| [112]     | R                  | Yes         | RD         | Late deliveries | Yes         | Multi-start Heuristic       | 1      |
| [113]     | R                  | No          | RD         | Delay           | Yes         | Brand-and-Benders-cut       | 1      |
| [114]     | R                  | Yes         | RD, DL     | Cost            | Yes         | Local search Heuristic      | 1      |
| [115]     | B                  | Yes         | RD         | Cost            | Yes         | GVNS                        | 1      |
| [116]     | B                  | Yes         | DL         | Route time      | Yes         | ALNS                        | k      |
| [117]     | B                  | Yes         | DL         | Route time      | Yes         | Two-phase Heuristic         | k      |
| [118]     | B                  | Yes         | DL         | -               | -           | Time, Cost Efficiency study | 1      |
| [119]     | B                  | Yes         | DL         | -               | -           | Energy Efficiency study     | 1      |
| [120]     | B                  | Yes         | CL         | Cost            | Yes         | Iterated Local Search       | 1      |
| [121]     | B (incl. Pick-ups) | Yes         | RD         | Cost            | Yes         | ALNS Heuristic              | k      |
| [122]     | R                  | No          | RD         | Robot depots    | Yes         | LNS Heuristic               | 1      |
| [123]     | R                  | No          | RD         | Tour Length     | No          | Simulation                  | 1      |
| [124]     | B                  | No          | RD         | Tour Length     | Yes         | Exact Algorithm\Simulation  | 1      |

- How can we leverage the communication capabilities of those IoT vehicles to continuously optimize the routing during operation based on their sensing abilities?
- How can the fleet of vehicles become a swarm of vehicles, working together, cooperatively in order to complete their operation optimally in an uncertain environment?
- How can we mitigate the increased sensitivity to externalities of these vehicles, and thus, the increased risk of failure for the operation, making the supply chain more robust?

The following subsections discuss potential future research directions and research topics that should be researched further.

## **4.1 Energy Consumption**

The energy consumption of battery EVs is the element with the highest effect on logistics operations, and it is the first and most important topic to address.

### **4.1.1 Payload Weight**

As discussed in the previous sections of this study, the payload of an EV greatly affects its range; however, the lack of real data means that approximations should be made for now. The advertised range of EV is that of the WLTP standard, at just 15% of the total payload. Aggressive reduction in range when EVs are loaded to capacity is a very realistic assumption to be made when heavy payloads are transported.

### **4.1.2 Vehicle Speed**

One of the least researched topics in EVRP is the effect of vehicle speed on energy consumption.

Additionally, different driving conditions, such as highway or city driving, can also have a significant impact on the actual range of an EV, which is not reflected in laboratory testing. Furthermore, more research is needed to understand the energy consumption of EVs over a longer period of time and under different driving conditions.

### **4.1.3 Stochastic Energy Consumption**

An additional step for realistic problem-solving in EVRP is the adoption of a stochastic energy consumption mechanism in all EVRPs. This addition will introduce a very important element of realism that is very representative of the real world and necessary for the development of robust solution methods. The stochastic consumption may represent extreme breaking incidents, extended use of auxiliaries, i.e., when being in a traffic jam, and other unpredictable events that do occur in daily

driving but cannot be modeled in another way. A way of implementing this stochastic consumption would be to assume an additional load over a unit of distance.

#### 4.1.4 Charging Curves

Besides energy consumption, EVRP is often concerned with the replenishment of the spent energy. Research in the literature has covered both battery swapping and battery charging. Unfortunately, the lack of standardization and the technical difficulties have not allowed battery swapping to become mainstream. Battery recharging has been the prevalent method used for all large vehicles, such as cars, vans, and trucks. The first EVRP papers that considered recharging assumed a linear charging function, meaning the charging time is proportional to the replenished energy. Later, non-linear charging functions were introduced as well. The charging behavior of some EVs can indeed be assumed to be linear such as Tesla vehicles; however, there are many more different types of charging curves as presented in Fig. 1. To date, there has been no research exploring a variety of real EV charging curves. Future research can include different charging profiles from existing EVs and compare them to one another to provide insights and find the EVs with the most desirable technological characteristics. This is essential as the optimal routes and charging stops could change significantly among different EVs.

#### 4.1.5 Multi-objective Models

Energy consumption has been the most significant difference introduced in VRP when using EVs. This has made energy consumption be treated as the single most important parameter. Nonetheless, there are many different applications that concurrently have additional objectives to fulfill. Multi-objective approaches have been presented in the EVRP literature; however, there are many opportunities for new research, especially when considering the new vehicle types presented in this study. Developing well-balanced multi-objective approaches is a challenging task; however, they are necessary for certain applications, i.e., when delivering perishable products with an EV, or when a drone is used in a medical emergency situation.

#### 4.1.6 New Benchmark Instances

All variants of EVRP to date have been using benchmark instances adapted from other VRP variants found in the literature. One of the first and necessary steps to have a better chance at depicting the real nature of EVs is to have a better representation of EVs when solving problems and benchmarking solution methods. The present study has provided the basis for the creation of new instances. While the distances to cover do not have to be realistic, using the energy capacity and energy consumption of existing EVs will lead to additional insights when solving these problems. Additionally, benchmark instances would have to be updated to represent real-world values, and energy consumption calculations should include the effects of payload weight.

## 4.2 New Types of Electric Vehicles

New benchmark instances are a step in the right direction; however, using data from a single EV will not provide the necessary level of realism. To go a step further, new types of EVs must be considered in EVRP.

### 4.2.1 Small Electric Delivery Vans

The new small-size electric vans presented in Subsection 2.4 have not been used in any EVRP variant so far. The use of smaller vehicles comes with unique challenges. The limited available space offers new research opportunities. EVRP can be extended to include multi-dimensional loading constraints to ensure maximum cargo space utilization. Research in multi-dimensional VRP variants has been very limited; however, it would vastly aid in maximizing the utilization rate of all-electric vans.

Small electric vans, like the ones portrayed in Fig. 3, have become the prevalent mean of urban deliveries in many European cities. Addressing their use can provide insights into their use and abilities.

### 4.2.2 Heterogeneous Fleets

Despite the widespread use of small EVs in Europe, each market and each business have different transportation needs. Using EVs of different sizes and capacities can provide logistics companies with more flexibility in their operations. Smaller EVs can be used for last-mile deliveries, while larger EVs can be used for longer hauls. Each company has different transportation needs. By using the right size of EV, logistics companies can improve their efficiency and reduce their costs.



**Fig. 3** Paxster EV — source: <https://paxster.no> (left), ALKE refrigerated EV — source: <https://www.alke.com> (right)

Smaller EVs can be used for smaller deliveries, reducing the need to use larger vehicles that may be underutilized. This can result in lower operational costs and reduced energy use. In addition, using smaller EVs for last-mile deliveries can help to reduce congestion in urban areas. These vehicles can be more maneuverable and able to navigate through congested streets more easily, reducing the time spent in traffic and improving delivery times.

Optimizing the fleet composition can have monetary benefits as well. First and foremost, battery size is directly related to the cost of the vehicle, as the batteries are generally the most costly component. Small EVs use small batteries and generally cost less. Regular-size electric vans come with batteries of different sizes. If a business has low range demand, they can select an electric van with a smaller battery and save on the initial cost of the fleet. Subsequently, the fleet composition can have a substantial effect on purchasing costs without sacrificing their operational capabilities.

Publications on the mixed-fleet EVRP have been limited. Researchers can enrich it by using vehicle data representative of the real world, with different charging curves for different types of vehicles and different battery sizes for the same vehicles. Furthermore, it is worth exploring different operational scenarios in regard to charging policies for each type of vehicle. Given the variety of EVs available today and the mentioned potential benefits, the use of heterogeneous EV fleets should be researched further to provide insights for real-world applications.

#### 4.2.3 Multi-echelon Approaches

One of the most efficient methods used to dissect the delivery operations when multiple delivery stages are involved is using multiple echelons. Traditionally, multi-echelon problems refer to scenarios where there are multiple levels of distribution centers or warehouses that need to be serviced by a fleet of vehicles. Frequently, each echelon has different vehicles. However, this scheme can be used to represent a system of interdependent EVs which coordinate with each other to achieve their goal. For example, large electric vans can be used for the transportation of goods, between facilities and make en route stops to serve customers or re-stock small electric vans or any other type of smart electric vehicle or station. The integration of various plans to develop intricate delivery systems presents significant research prospects for enhancing overall delivery efficiency in a controlled manner. Most multi-echelon approaches in VRP and EVRP consist of two echelons. Future research can focus on adding more echelons and present multi-objective models with different objectives in each echelon.

#### 4.2.4 Drones and Robots

Drones and robots have created the opportunity for faster and greener transportation, especially when combined with electric vans. As discussed in the previous sections of this study, they can offer great operational agility and savings. Drones in VRP have been researched more, compared to robots; however, only one study exists on



the EVRPD and none on the EVRP with robots (EVRPR). This means that there are many opportunities for research in both.

The proposed integration presents a highly valuable area for future research, offering great potential for reducing energy consumption and improving delivery efficiency. Besides the obvious extensions of VRPD variants in order to include EVs, new applications can be developed. Drones can be used to visit automated lockers at known positions, given that lockers have become more popular in recent years. Subsequently, the potential places to visit would remain generally unchanged allowing for a more accurate prediction of energy needs. Drones can also be used for pickup operations, with the objective of minimizing the total energy consumption of the electric van they serve. In heterogeneous fleet scenarios, drones can be used to transfer items between vehicles whenever necessary.

Robots have in general attracted less attention in VRP and have been mainly proposed for use in warehouses and factories. One new scenario to explore with robots in combination with electric vans is their use as mobile lockers for small parcels. Such a combination would lower the total operational time as it would limit the number of times that a driver has to find a parking space and make the delivery. Robots could also be used for trash collection. Instead of having a large EV visit each site in an urban center, robots could be used to eliminate some of the stops, especially in urban city centers with lots of road traffic.

The potential variants to be developed are plenty, with many new insights to be presented, combining elements from different works found in the literature and adding new parameters in an attempt to have a more realistic representation. However, the concurrent use of multiple vehicle types makes the network more complex and introduces new points of failure. In conjunction with the strong coupling between them, the impact of operational failures can be more noticeable compared to the conventional VRPs. To overcome potential setbacks, future research should focus on generating flexible delivery plans with a reasonable level of redundancy.

In addition, while all vehicles operate within the same environment and are affected by the same parameters, the significance level of each parameter to each vehicle type can be different. Therefore, it is important to determine the sensitivity of each type of vehicle to each parameter and evaluate the level of influence it should have. Moreover, some parameters may be interdependent, i.e., a strong headwind will have a toll on the energy consumption of a vehicle traveling at highway speeds, but it would not make a difference in the center of a city.

### 4.3 New Solution Methods

EVs are both a mode of transportation and a technological product. They are equipped with a variety of sensors and electronic components that generate useful data, such as battery charge levels, charging rates, and energy usage, which can be analyzed to optimize battery life and charging patterns. Additionally, EVs equipped with advanced sensors and driver assistance features generate data on driving behavior and road conditions, which can be used to improve safety and enhance driving efficiency. These

features are of high value for logistics operators as that data can be used to assess and improve their operations.

To make such information useful, it is necessary to develop simulation tools that can help recreate different scenarios, evaluate past decisions, and create robust routing plans. The introduction of stochastic energy consumption and the development of data-driven approaches are a great combination for carrying out simulations. By running multiple simulations and analyzing the results, the likelihood of certain events occurring can be estimated, such as a vehicle running out of battery before reaching a charging station. In addition, researchers can simulate different traffic patterns, battery capacities, and charging station locations to see how they affect optimal routing solutions.

A tool that could be used for simulations is artificial intelligence (AI), which has become so popular due to its ability to process and analyze large amounts of data, automate repetitive tasks, improve decision-making, and develop solutions to complex problems. While AI methods have been used in the past to solve VRP problems, there is the opportunity of using AI methods in EVRP simulations. Two methods that could be employed in EVRP are machine learning (ML) algorithms and reinforcement learning (RL) algorithms. ML algorithms can be used to analyze and learn from large amounts of data, making it possible to generate more accurate and efficient EVRP solutions. For example, they can be used to predict traffic patterns and weather conditions, allowing for more accurate route planning. RL algorithms can be used in case data is not available, as RL implementations use the process of trial and error to determine how to make good choices. RL algorithms can be used to continuously improve routing decisions over time by learning from past experiences.

This creates a research opportunity for the development of simulation tools that will help decision-makers determine their EV needs and provide them with adequate knowledge to make informed decisions. Simulations will also give insights in the reliability of EVs for the operations considered.

Furthermore, unlike the conventional VRPs with stochastic environmental variables, such as stochastic travel times and stochastic demand, uncertainty in these novel routing problems with electric vehicles has a pivotal role, as it can disrupt the whole routing operation. For example, uncertainty in payload weight, traffic, or weather conditions can significantly alter the range of operation of an electric vehicle. Combined with uncertainty in the availability of charging stations can be detrimental even to the best-optimized routing plan. Charging stations may be full, not have the available charger, suffer from a temporary failure, or have other issues, as explained in the previous sections. Simulations can help assess the reliability of EVs by exploring different scenarios of charger availability and malfunctions.

Another potential use of such simulations would be their use for the assessment of the routing plans provided by the deterministic heuristic and meta-heuristic algorithms that have been prevalent in VRP research.

#### 4.4 Communication

The novel vehicle types discussed previously are de facto IoT devices, as they incorporate sensors and they have the ability to process data and communicate with each other or other devices. Until the late 2000s and the emergence of the smartphone as the first mobile device connected to the internet, the concept of interconnected autonomous vehicles was far-fetched. Today, autonomous driving vehicles are a technological reality.

These IoT vehicles are able to exchange information instantly, in real time, and thus, have the potential to cooperatively overcome unpredictable events, which otherwise would be detrimental. This adaptive ability of the vehicles, as a swarm, can offer practical benefits which would be unattainable 10 years ago.

Vehicle-to-vehicle communication, referred to as V2V, allows vehicles to exchange information with each other in real time, such as their location, speed, and route. This information can be used to optimize the routing of vehicles in a fleet. For example, if one vehicle in a fleet encounters unexpected traffic or road closures, it can quickly relay this information to the rest of the fleet, allowing them to re-route from an earlier time, instead of reactively following a new route.

Research regarding VRPs should consider leveraging these capabilities in order to not only optimize a static unrealistic version of the problem at hand but also, dynamic and uncertain versions of it. The insights regarding the impact of the adaptive behavior of the vehicle fleets, as well as the coordinating strategies themselves, will be useful from both a theoretical standpoint and in practical applications. Communication between vehicles is an element that is not addressed in existing literature, while it is essential to revolutionizing supply chains.

### 5 Conclusions

Electric vehicles, such as electric vans, drones, and robots, are very promising, novel means of transportation. This paper presented the technical characteristics along with the advantages they offer and the limitations they impose. Furthermore, the existing literature on VRPs employing electric vehicles, drones, robots, and their possible combinations was reviewed and discussed in terms of realism and applicability, showcasing a research gap between the literature and practical applications.

Sustainability in the supply chain and logistics operations is a critical issue that requires a step-by-step approach to ensure its success. From the perspective of logistics, sustainability in the supply chain and logistics operations refers to the management of environmental, social, and economic impacts throughout the life cycle of goods and materials. This includes the efficient and sustainable transportation of goods, responsible and efficient inventory and materials management, and the promotion of sustainable practices and behaviors among employees, suppliers, and other stakeholders in the supply chain.

The publications presented and the concepts they have introduced have greatly contributed to advancing the VRP field of study. All researchers proposing VRP models utilizing novel means of transportation have provided useful insights on

their potential impact. As we move forward, almost 65 years after the VRP was introduced, we should aim to move away from abstracted, static approaches of real-life applications. Research should embrace dynamic environments, stochasticity, and the complexity encountered in practical supply chains, as it is the only way to truly explore the capabilities of those novel means of transportation combined.

In order to make the leap to the next-generation VRPs, based on the discussed publications, the capabilities of the state-of-the-art means of transportation, and the requirements of the modern supply chains for sustainability and autonomy, we proposed the establishment of a common ground for defining and studying novel VRPs and testing solution approaches.

In summary, the proposed elements for the next generations of VRP to consider were the following:

- **Energy consumption:** The topic of energy consumption can provide many opportunities for future research. The use of parameters such as payload and speed should become a part of all energy consumption functions. Stochastic consumption elements should be used to represent energy spent unpredictably. Charging functions, linear and non-linear, should be updated to represent real-world charging functions and explore different strategies for different types of vehicles. In addition, new benchmark instances should be introduced that are closer to reality, thus, providing results that are easier to assess and distinguish from each other, instead of presenting insights based on arbitrary numbers and tests that do not represent reality.
- **New vehicle types:** In the past decade, many new EVs have been presented. Established manufacturers have offered electric variants of their existing EVs, but there are many new manufacturers that have emerged, some of which have created small electric vans which have become very popular. Other types of EVs that have become very popular in the past decade are drones and robots. Future research can focus on maximizing the use of small electric vans and incorporating them into logistics operations that make use of them. Researchers can also focus on the integration of drones and robots in logistics operations with EVs. Heterogeneous fleet compositions should also be studied more thoroughly and EV-specific approaches could be presented.
- **Dynamic environment:** The environment of logistics operations is never static in practice, as it contains unpredictable events and states. The computational capabilities of today make it easy to include dynamic elements and essentially simulate operations in a realistic manner. EVs generate useful data that can be analyzed to optimize their use. This data can be used in simulations that can help operators assess their reliability and foresee potential shortcomings. In addition, simulations can be a useful tool for assessing the feasibility of solutions proposed by deterministic approaches. Future research can focus on the development of simulation tools and transition from deterministic solutions to dynamic ones.
- **Communication:** Vehicles nowadays are IoT devices; therefore, they should communicate, exchange information, and leverage that capability to adapt in real time to changes, such as delays, weather, or other externalities. Contingency plans also have to be developed.

Based on these directions, research on VRP will answer many questions regarding the optimal integration of these novel vehicles, will explore their full potential, and will provide more accurate insights with respect to their true capabilities.

**Funding** Open access funding provided by HEAL-Link Greece.



research work was supported by the Hellenic Foundation for Research and Innovation (HFRI) under the HFRI PhD Fellowship grant (Fellowship Number: 6334.)

**Data Availability** Not applicable.

## Declarations

**Conflict of Interest** The authors declare no competing interests.

**Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

## References

1. Dantzig GB, Ramser JH (1959) The truck dispatching problem. *Manage Sci* 6(1):80–91
2. Kumar SN, Panneerselvam R (2012) A survey on the vehicle routing problem and its variants. *Intell Inf Manag* 4(3):19355
3. Helgeson B, Peter J (2020) The role of electricity in decarbonizing european road transport-development and assessment of an integrated multi-sectoral model. *Appl Energy* 262:114365
4. Schneider M, Stenger A, Goeke D (2014) The electric vehicle-routing problem with time windows and recharging stations. *Transp Sci* 48(4):500–520
5. Xiao Y, Zhang Y, Kaku I, Kang R, Pan X (2021) Electric vehicle routing problem: a systematic review and a new comprehensive model with nonlinear energy recharging and consumption. *Renew Sustain Energy Rev* 151:111567
6. Karakostas P, Sifaleras A (2022) Recent trends in sustainable supply-chain optimization. In: Fathi M, Zio E, Pardalos PM (eds) *Handbook of smart energy systems*. Springer
7. Patella SM, Grazieschi G, Gatta V, Marcucci E, Carrese S (2020) The adoption of green vehicles in last mile logistics: a systematic review. *Sustainability* 13(1):6
8. Hossain M, Kumar L, Islam M, Selvaraj J et al (2022) A comprehensive review on the integration of electric vehicles for sustainable development. *J Adv Transp* 2022
9. Goodchild A, Toy J (2018) Delivery by drone: an evaluation of unmanned aerial vehicle technology in reducing CO2 emissions in the delivery service industry. *Transp Res Part D: Transp Environ* 61:58–67
10. Pugliese LDP, Guerriero F, Macrina G (2020) Using drones for parcels delivery process. *Procedia Manuf* 42:488–497
11. Campbell JF (2022) Will drones revolutionize home delivery? Let's get real.... *Patterns* 3(8):100564

12. Peter O, Pradhan A, Mbohwa C (2023) Industrial internet of things (IIoT): opportunities, challenges, and requirements in manufacturing businesses in emerging economies. *Procedia Comput Sci* 217:856–865
13. Psannis KE, Xinogalos S, Sifaleras A (2014) Convergence of internet of things and mobile cloud computing. *Syst Sci Control Eng: An Open Access Journal* 2(1):476–483
14. Pourrahmani H, Yavarinasab A, Zahedi R, Gharehghani A, Mohammadi MH, Bastani P et al (2022) The applications of internet of things in the automotive industry: a review of the batteries, fuel cells, and engines. *Internet of Things* 100579
15. Qin H, Su X, Ren T, Luo Z (2021) A review on the electric vehicle routing problems: variants and algorithms. *Front Eng Manag* 8(3):370–389
16. Quak H, Nesterova N, van Rooijen T (2016) Possibilities and barriers for using electric-powered vehicles in city logistics practice. *Transp Res Proc* 12:157–169
17. Juan AA, Mendez CA, Faulin J, De Armas J, Grasman SE (2016) Electric vehicles in logistics and transportation: a survey on emerging environmental, strategic, and operational challenges. *Energies* 9(2):86
18. Anosike A, Loomes H, Udokporo CK, Garza-Reyes JA (2021) Exploring the challenges of electric vehicle adoption in final mile parcel delivery. *Int J Log Res Appl* 1–25
19. Rojas Vitoria D, Solano-Charris EL, Muñoz-Villamizar A, Montoya-Torres JR (2021) Unmanned aerial vehicles/drones in vehicle routing problems: a literature review. *Int Trans Oper Res* 28(4):1626–1657
20. Kellermann R, Biehle T, Fischer L (2020) Drones for parcel and passenger transportation: a literature review. *Transp Res Interdiscip Perspect* 4:100088
21. Narayanan S, Chaniotakis E, Antoniou C (2020) Shared autonomous vehicle services: a comprehensive review. *Transp Res Part C Emerg Technol* 111:255–293
22. Golbabaei F, Yigitcanlar T, Bunker J (2021) The role of shared autonomous vehicle systems in delivering smart urban mobility: a systematic review of the literature. *Int J Sustain Transp* 15(10):731–748
23. Fragapane G, De Koster R, Sgarbossa F, Strandhagen JO (2021) Planning and control of autonomous mobile robots for intralogistics: literature review and research agenda. *Eur J Oper Res* 294(2):405–426
24. Li H, Chen J, Wang F, Bai M (2021) Ground-vehicle and unmanned-aerial-vehicle routing problems from two-echelon scheme perspective: a review. *Eur J Oper Res* 294(3):1078–1095
25. Bahrami S, Nourinejad M, Amirjamshidi G, Roorda MJ (2020) The plugin hybrid electric vehicle routing problem: a power-management strategy model. *Transp Res Part C Emerg Technol* 111:318–333
26. Subramanyam A, Cokyasar T, Larson J, Stinson M (2022) Joint routing of conventional and range-extended electric vehicles in a large metropolitan network. *Transp Res Part C Emerg Technol* 144:103830
27. Kirsch DA (2000) *The electric vehicle and the burden of history*. Rutgers University Press. <https://books.google.gr/books?id=EqUQAQAAMAAJ>
28. Alizon F, Shooter SB, Simpson TW (2008) Henry ford and the model T: lessons for product platforming and mass customization. In: *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, vol 43291. pp 59–66
29. Thomas V, Maine E (2019) Market entry strategies for electric vehicle start-ups in the automotive industry-lessons from Tesla Motors. *J Clean Prod* 235:653–663
30. Hoekstra A (2019) The underestimated potential of battery electric vehicles to reduce emissions. *Joule* 3(6):1412–1414
31. Hawkins TR, Singh B, Majeau-Bettez G, Strømman AH (2013) Comparative environmental life cycle assessment of conventional and electric vehicles. *J Ind Ecol* 17(1):53–64
32. Verma S, Dwivedi G, Verma P (2022) Life cycle assessment of electric vehicles in comparison to combustion engine vehicles: a review. *Mater Today: Proc* 49:217–222
33. Mpoi G, Milioti C, Mitropoulos L (2023) Factors and incentives that affect electric vehicle adoption in Greece. *Int J Transp Sci Technol*
34. Mercedes-Benz. Average daily trips. <https://www.mercedes-benz.co.uk/vans/en/e-vito-panel-van>. Accessed 15 Mar 2022
35. Kyriakakis NA, Stamadianos T, Marinaki M, Marinakis Y (2022) The electric vehicle routing problem with drones: an energy minimization approach for aerial deliveries. *Cleaner Logistics and Supply Chain* 100041

36. Savari GF, Sathik MJ, Raman LA, El-Shahat A, Hasanien HM, Almakhlles D, Aleem SHA, Omar AI (2022) Assessment of charging technologies, infrastructure and charging station recommendation schemes of electric vehicles: a review. *Ain Shams Eng J* 101938
37. Goetzel N, Hasanuzzaman M (2022) An empirical analysis of electric vehicle cost trends: a case study in Germany. *Res Transp Bus Manag* 43:100825
38. Liu K, Wang J, Yamamoto T, Morikawa T (2018) Exploring the interactive effects of ambient temperature and vehicle auxiliary loads on electric vehicle energy consumption. *Appl Energy* 227:324–331
39. Ma Y, Ding H, Liu Y, Gao J (2022) Battery thermal management of intelligent-connected electric vehicles at low temperature based on NMPC. *Energy* 244:122571
40. Al-Wreikat Y, Serrano C, Sodr  JR (2022) Effects of ambient temperature and trip characteristics on the energy consumption of an electric vehicle. *Energy* 238:122028
41. Hato Y, Chen CH, Hirota T, Kamiya Y, Daisho Y, Inami S (2015) Degradation predictions of lithium iron phosphate battery. *World Electr Veh J* 7(1):25–31
42. Marra F, Yang GY, Tr holt C, Larsen E, Rasmussen CN, You S (2012) Demand profile study of battery electric vehicle under different charging options. In: 2012 IEEE Power and Energy Society General Meeting. IEEE, pp 1–7
43. Kostopoulos ED, Spyropoulos GC, Kaldellis JK (2020) Real-world study for the optimal charging of electric vehicles. *Energy Rep* 6:418–426
44. Lin J, Zhou W, Wolfson O (2016) Electric vehicle routing problem. *Transp Res Proc* 12:508–521
45. Shao S, Guan W, Bi J (2017) Electric vehicle-routing problem with charging demands and energy consumption. *IET Intel Transport Syst* 12(3):202–212
46. Xiao Y, Zuo X, Kaku I, Zhou S, Pan X (2019) Development of energy consumption optimization model for the electric vehicle routing problem with time windows. *J Clean Prod* 225:647–663
47. Zhang S, Gajpal Y, Appadoo SS, Abdulkader MMS (2018) Electric vehicle routing problem with recharging stations for minimizing energy consumption. *Int J Prod Econ* 203:404–413
48. Basso R, Kulcs r B, Egd rt B, Lindroth P, Sanchez-Diaz I (2019) Energy consumption estimation integrated into the electric vehicle routing problem. *Transp Res Part D: Transp Environ* 69:141–167
49. Rastani S, Y ksel T,  atay B (2019) Effects of ambient temperature on the route planning of electric freight vehicles. *Transp Res Part D: Transp Environ* 74:124–141
50. Cokyasar T, Subramanyam A, Larson J, Stinson M, Sahin O (2023) Time-constrained capacitated vehicle routing problem in urban e-commerce delivery. *Transp Res Rec* 2677(2):190–203
51. Veneri O, Ferraro L, Capasso C, Iannuzzi D (2012) Charging infrastructures for EV: overview of technologies and issues. In: 2012 Electrical Systems for Aircraft, Railway and Ship Propulsion. IEEE, pp 1–6
52. Jia Q-S, Long T (2020) A review on charging behavior of electric vehicles: data, model, and control. *Control Theory Technol* 18(3):217–230
53. Desaulniers G, Errico F, Irnich S, Schneider M (2016) Exact algorithms for electric vehicle-routing problems with time windows. *Oper Res* 64(6):1388–1405
54. Ferro G, Paolucci M, Robba M (2018) An optimization model for electrical vehicles routing with time of use energy pricing and partial recharging. *IFAC-PapersOnLine* 51(9):212–217
55. Bruglieri M, Mancini S, Pezzella F, Pisacane O, Suraci S (2017) A three-phase matheuristic for the time-effective electric vehicle routing problem with partial recharges. *Electron Notes Discrete Math* 58:95–102
56. Ding N, Batta R, Kwon C et al (2015) Conflict-free electric vehicle routing problem with capacitated charging stations and partial recharge. In: Technical Report. Department of Industrial and Systems Engineering, University at Buffalo, US
57.  atay B, Keskin M (2017) The impact of quick charging stations on the route planning of electric vehicles. In: 2017 IEEE Symposium on Computers and Communications (ISCC). IEEE, pp 152–157
58. Zuo X, Xiao Y, You M, Kaku I, Xu Y (2019) A new formulation of the electric vehicle routing problem with time windows considering concave nonlinear charging function. *J Clean Prod* 236:117687
59. Froger A, Mendoza JE, Jabali O, Laporte G (2017) A matheuristic for the electric vehicle routing problem with capacitated charging stations
60. Karakati  S (2021) Optimizing nonlinear charging times of electric vehicle routing with genetic algorithm. *Expert Syst Appl* 164:114039

61. Zang Y, Wang M, Qi M (2022) A column generation tailored to electric vehicle routing problem with nonlinear battery depreciation. *Comput Oper Res* 137:105527
62. Botsford C, Szczepanek A (2009) Fast charging vs. slow charging: pros and cons for the new age of electric vehicles. In: *International Battery Hybrid Fuel Cell Electric Vehicle Symposium*. Citeseer, pp 1–9
63. Li Y, Li K, Xie Y, Liu J, Fu C, Liu B (2020) Optimized charging of lithium-ion battery for electric vehicles: adaptive multistage constant current-constant voltage charging strategy. *Renew Energy* 146:2688–2699
64. Keskin M, Çatay B (2018) A matheuristic method for the electric vehicle routing problem with time windows and fast chargers. *Comput Oper Res* 100:172–188
65. Froger A, Mendoza JE, Jabali O, Laporte G (2019) Improved formulations and algorithmic components for the electric vehicle routing problem with nonlinear charging functions. *Comput Oper Res* 104:256–294
66. Raeesi R, Zografos KG (2020) The electric vehicle routing problem with time windows and synchronised mobile battery swapping. *Transp Res B Methodol* 140:101–129
67. Jie W, Yang J, Zhang M, Huang Y (2019) The two-echelon capacitated electric vehicle routing problem with battery swapping stations: formulation and efficient methodology. *Eur J Oper Res* 272(3):879–904
68. Raeesi R, Zografos KG (2022) Coordinated routing of electric commercial vehicles with intra-route recharging and en-route battery swapping. *Eur J Oper Res* 301(1):82–109
69. Ogura K, Kolhe ML (2017) Battery technologies for electric vehicles. In: *Electric vehicles: prospects and challenges*. Elsevier, pp 139–167
70. Hjelkrem OA, Arnesen P, Bø TA, Sondell RS (2020) Estimation of tank-to-wheel efficiency functions based on type approval data. *Appl Energy* 276:115463
71. Hõimoja H, Rufer A, Dziechciaruk G, Vezzini A (2012) An ultrafast EV charging station demonstrator. In: *International Symposium on Power Electronics Power Electronics, Electrical Drives, Automation and Motion*. IEEE, pp 1390–1395
72. WLTP. [https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:02017R1151-20190101 & from=EN](https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:02017R1151-20190101&from=EN). Accessed 7 Feb 2023
73. Jung S, Kim H (2017) Analysis of amazon prime air UAV delivery service. *Journal of Knowledge Information Technology and Systems* 12(2):253–266
74. DHL Parcel-copter. <https://www.dhl.com/discover/en-global/business/business-ethics/parcelcopter-drone-technology>. Accessed 1 Apr 2022
75. Matternet M2. <https://mttr.net/product>. Accessed 1 Apr 2022
76. Zipline Drone. [https://en.wikipedia.org/wiki/Zipline\\_\(drone\\_delivery\)](https://en.wikipedia.org/wiki/Zipline_(drone_delivery)). Accessed 1 Apr 2022
77. Rodrigues TA, Patrikar J, Oliveira NL, Matthews HS, Scherer S, Samaras C (2022) Drone flight data reveal energy and greenhouse gas emissions savings for very small package delivery. *Patterns* 3(8):100569
78. Murray CC, Chu AG (2015) The flying sidekick traveling salesman problem: optimization of drone-assisted parcel delivery. *Transp Res Part C Emerg Technol* 54:86–109
79. Kitjacharoenchai P, Min B-C, Lee S (2020) Two echelon vehicle routing problem with drones in last mile delivery. *Int J Prod Econ* 225:107598
80. Cengiz E, Yilmaz C, Kahraman HT, Suiçmez Ç (2023) Effects of variable UAV speed on optimization of travelling salesman problem with drone (TSP-D). In: *Smart applications with advanced machine learning and human-centred problem design*. Springer, pp 295–305
81. Raj R, Murray C (2020) The multiple flying sidekicks traveling salesman problem with variable drone speeds. *Transp Res Part C Emerg Technol* 120:102813
82. Wang Z, Sheu J-B (2019) Vehicle routing problem with drones. *Transport Res B-Meth* 122:350–364
83. Chauhan D, Unnikrishnan A, Figliozzi M (2019) Maximum coverage capacitated facility location problem with range constrained drones. *Transp Res Part C Emerg Technol* 99:1–18
84. Coindreau M-A, Gallay O, Zufferey N (2019) Vehicle routing with transportable resources: using carpooling and walking for on-site services. *Eur J Oper Res* 279(3):996–1010
85. Gu Q, Fan T, Pan F, Zhang C (2020) A vehicle-UAV operation scheme for instant delivery. *Comput Ind Eng* 149:106809
86. Hu M, Liu W, Lu J, Fu R, Peng K, Ma X, Liu J (2019) On the joint design of routing and scheduling for vehicle-assisted multi-UAV inspection. *Futur Gener Comput Syst* 94:214–223
87. Deng P, Amirjamshidi G, Roorda M (2020) A vehicle routing problem with movement synchronization of drones, sidewalk robots, or foot-walkers. *Transp Res Proc* 46:29–36



88. Karak A, Abdelghany K (2019) The hybrid vehicle-drone routing problem for pick-up and delivery services. *Transp Res Part C Emerg Technol* 102:427–449
89. Kitjacharoenchai P, Ventresca M, Moshref-Javadi M, Lee S, Tanchoco JM, Brunese PA (2019) Multiple traveling salesman problem with drones: mathematical model and heuristic approach. *Comput Ind Eng* 129:14–30
90. Rossello NB, Garone E (2020) Carrier-vehicle system for delivery in city environments. *IFAC-PapersOnLine* 53(2):15253–15258
91. Li H, Wang H, Chen J, Bai M (2020) Two-echelon vehicle routing problem with time windows and mobile satellites. *Transport Res B-Meth* 138:179–201
92. Nguyen MA, Dang GT-H, Hà MH, Pham M-T (2022) The min-cost parallel drone scheduling vehicle routing problem. *Eur J Oper Res* 299(3):910–930
93. Zhen L, Li M, Laporte G, Wang W (2019) A vehicle routing problem arising in unmanned aerial monitoring. *Comput Oper Res* 105:1–11
94. Shahzaad B, Bouguettaya A, Mistry S, Neiat AG (2021) Resilient composition of drone services for delivery. *Futur Gener Comput Syst* 115:335–350
95. Thibbotuwawa A, Bocewicz G, Nielsen P, Zbigniew B (2019) Planning deliveries with UAY routing under weather forecast and energy consumption constraints. *IFAC-PapersOnLine* 52(13):820–825
96. Liu Y (2019) An optimization-driven dynamic vehicle routing algorithm for on-demand meal delivery using drones. *Comput Oper Res* 111:1–20
97. Macias JE, Angeloudis P, Ochieng W (2020) Optimal hub selection for rapid medical deliveries using unmanned aerial vehicles. *Transp Res Part C Emerg Technol* 110:56–80
98. Cheng C, Adulyasak Y, Rousseau L-M (2020) Drone routing with energy function: formulation and exact algorithm. *Transport Res B-Meth* 139:364–387
99. Vidal T, Laporte G, Matl P (2020) A concise guide to existing and emerging vehicle routing problem variants. *Eur J Oper Res* 286(2):401–416
100. Macrina G, Pugliese LDP, Guerriero F, Laporte G (2020) Drone-aided routing: a literature review. *Transp Res Part C Emerg Technol* 120:102762
101. Cheikhrouhou O, Khoufi I (2021) A comprehensive survey on the multiple traveling salesman problem: applications, approaches and taxonomy. *Comput Sci Rev* 40:100369
102. Moshref-Javadi M, Winkenbach M (2021) Applications and research avenues for drone-based models in logistics: a classification and review. *Expert Syst Appl* 177:114854
103. Chung SH, Sah B, Lee J (2020) Optimization for drone and drone-truck combined operations: a review of the state of the art and future directions. *Comput Oper Res* 123:105004
104. Amazon Scout. <https://www.dimensions.com/element/amazon-scout>. Accessed 1 Apr 2022
105. Starship Robot. <https://www.post.ch/-/media/post/ueber-uns/medienmitteilungen/2017/factsheet-lieferroboter.pdf?la=en>. Accessed 1 Apr 2022
106. Teleretail Delivery Robot. <https://www.dimensions.com/element/teleretail-delivery-robot>. Accessed 1 Apr 2022
107. Deliro Delivery Robot. <https://www.dimensions.com/element/deliro-delivery-robot>. Accessed 1 Apr 2022
108. Kiwibot. <https://www.dimensions.com/element/kiwibot>. Accessed 1 Apr 2022
109. Serve Robotics. <https://www.dimensions.com/element/serve-serve-robotics>. Accessed 1 Apr 2022
110. Fed-Ex Roxo. <https://www.dimensions.com/element/fedex-roxo>. Accessed 1 Apr 2022
111. Nuro R2. <https://www.dimensions.com/element/nuro-r2>. Accessed 1 Apr 2022
112. Boysen N, Schwedfeger S, Weidinger F (2018) Scheduling last-mile deliveries with truck-based autonomous robots. *Eur J Oper Res* 271(3):1085–1099
113. Alfandari L, Ljubić I, da Silva MDM (2022) A tailored benders decomposition approach for last-mile delivery with autonomous robots. *Eur J Oper Res* 299(2):510–525
114. Ostermeier M, Heimfarth A, Hübner A (2022) Cost-optimal truck-and-robot routing for last-mile delivery. *Networks* 79(3):364–389
115. Heimfarth A, Ostermeier M, Hübner A (2022) A mixed truck and robot delivery approach for the daily supply of customers. *Eur J Oper Res* 303(1):401–421
116. Chen C, Demir E, Huang Y (2021) An adaptive large neighborhood search heuristic for the vehicle routing problem with time windows and delivery robots. *Eur J Oper Res* 294(3):1164–1180
117. Chen C, Demir E, Huang Y, Qiu R (2021) The adoption of self-driving delivery robots in last mile logistics. *Transp Res E: Logist Transp Rev* 146:102214
118. Jennings D, Figliozzi M (2019) Study of sidewalk autonomous delivery robots and their potential impacts on freight efficiency and travel. *Transp Res Rec* 2673(6):317–326

119. Figliozzi M, Jennings D (2020) Autonomous delivery robots and their potential impacts on urban freight energy consumption and emissions. *Transp Res Proc* 46:21–28
120. Simoni MD, Kutanoglu E, Claudel CG (2020) Optimization and analysis of a robot-assisted last mile delivery system. *Transp Res E: Logist Transp Rev* 142:102049
121. Yu S, Puchinger J, Sun S (2022) Van-based robot hybrid pickup and delivery routing problem. *Eur J Oper Res* 298(3):894–914
122. Bakach I, Campbell AM, Ehmke JF (2021) A two-tier urban delivery network with robot-based deliveries. *Networks* 78(4):461–483
123. Poeting M, Schaudt S, Clausen U (2019) A comprehensive case study in last-mile delivery concepts for parcel robots. In: 2019 Winter Simulation Conference (WSC). IEEE, pp 1779–1788
124. Poeting M, Schaudt S, Clausen U (2019) Simulation of an optimized last-mile parcel delivery network involving delivery robots. In: Interdisciplinary Conference on Production, Logistics and Traffic. Springer, pp 1–19

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.